Solving systems of equations with a variety of methods

1 Describe the graphical solution of a system of two linear equations.

2 Find the number that should be multiplied to the second equation of the following system of equations to eliminate y using addition method:

$$\begin{cases} -3x - 12y = 6 \\ -2x - 4y = -4 \end{cases}$$

3 If the first equation is multiplied by 3, find the number that should be multiplied to the second equation of the following system of equations to eliminate y using addition method:

$$\begin{cases} 4x + 4y = 6 \\ 5x + 3y = 3 \end{cases}$$

4 Consider the equations $y = 3x - 1$ and $y = -5x + 23$.

a. Complete the given table by finding the y-values for each of the given x-values:

b. Hence, find the values for x and y which satisfy both the equations.

x	$y = 3x - 1$	$y = -5x + 23$
1		
2		
3		
4		
5		

5 Consider the equations $y = 5x$ and $y = 9 - 4x$.

a. Complete the given table by finding the y-values for each of the given x-values:

b. Hence, find the values for x and y which satisfy both the equations.

x	$y = 5x$	$y = 9 - 4x$
1		
3		
5		

6 Consider the following linear equations:

$$y=\frac{x}{3}+\frac{1}{3} \text{ and } -8y=8x+8$$

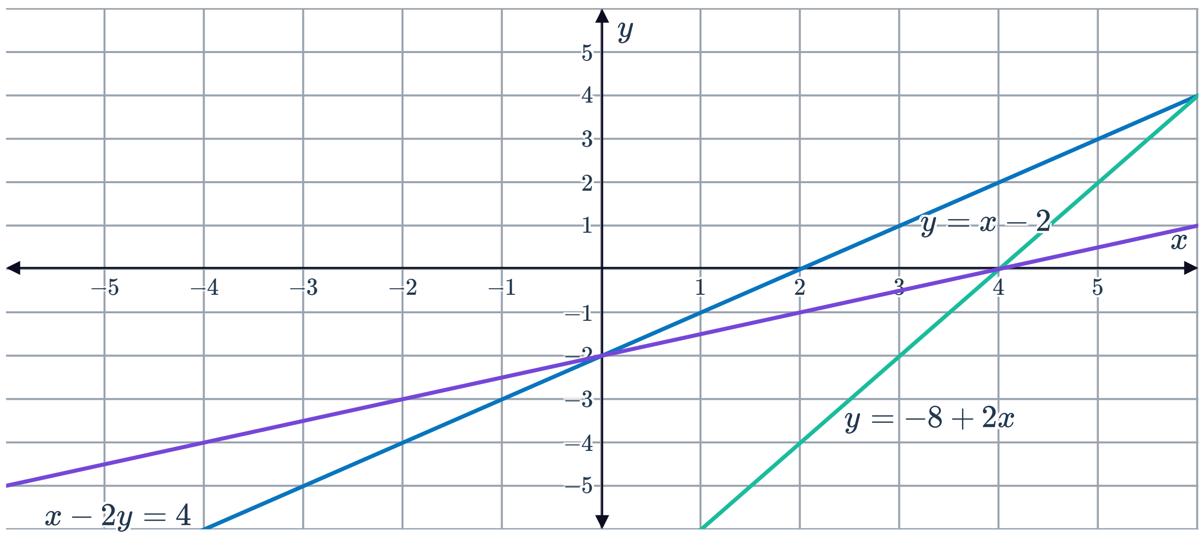
a Find the intercepts of $y=\dfrac{x}{3}+\dfrac{1}{3}$.

b Find the intercepts of $-8y=8x+8$.

c Sketch the lines of the two equations on the same coordinate plane.

d Hence, state the values of x and y which satisfy both equations.

7 Use the given graph to solve each system of equations:



a $$\begin{aligned} y&=x-2 \\ y&=-8+2x \end{aligned}$$

b $$\begin{aligned} y&=x-2 \\ x-2y&=4 \end{aligned}$$

c $$\begin{aligned} y&=-8+2x \\ x-2y&=4 \end{aligned}$$

8 Solve the following systems of equations:

a $$\begin{aligned} 2x-5y&=1 \\ -3x-5y&=-39 \end{aligned}$$

b $$\begin{aligned} 2x-5y&=-10 \\ y&=-5x+6 \end{aligned}$$

c $$\begin{aligned} -\frac{x}{4}+\frac{y}{5}&=8 \\ \frac{x}{5}+\frac{y}{3}&=1 \end{aligned}$$

d $$\begin{aligned} 5x+3y&=15 \\ \frac{6y}{5}&=10-4x \end{aligned}$$

e $$\begin{aligned} -7p+2q&=-\frac{13}{10} \\ -21p+10q&=-\frac{9}{10} \end{aligned}$$

f $$\begin{aligned} -5p-7q&=-\frac{43}{5} \\ -15p-28q&=-\frac{157}{5} \end{aligned}$$

9 Consider the following linear equations:

$$
\begin{aligned}
-6x - 2y &= -28 \\
2x + 16y &= 40 \\
4x - 2y &= 12
\end{aligned}
$$

a Find the values of x and y that satisfy the first two equations.

b Determine if this solution satisfies the third equation by substituting the values of x and y into the left hand side of the equation.

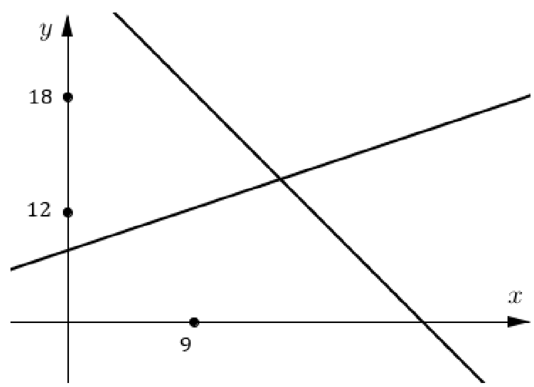
c Hence, state whether the lines are concurrent.

10 Suppose two numbers add up to 9 and have a difference of 1. Let x represent the smaller number and let y represent the other number. Write a system of equations that describes this problem.

11 Marge graphed a system of two linear equations. She began with an approximation of $x = 9$ and it did not satisfy both equations simultaneously. One equation gave a resulting value of $y = 12$ and the other equation gave a resulting value of $y = 18$ as shown. Note that the graph is not to scale.

a State the closer approximation to the solution: $x = 5$ or $x = 13$.

b She substituted $x = 13$ into both equations and found that it did not satisfy the two equations simultaneously. She also found that the difference between the resulting y values was 10.
State the x value closest to the solution: $x = 9$ or $x = 13$.



12 Consider the system of linear equations.

$$\text{Equation 1: } y = x - 6$$
$$\text{Equation 2: } y = 5x + 6$$

a The following system of equations was obtained by adding a multiple of Equation 2 to Equation 1.

$$\text{Equation 3: } y = x - 6$$
$$\text{Equation 4: } 5y = 21x + 18$$

Find the multiple of Equation 2 that was added.

b Find the values of x and y that satisfy Equation 3 and Equation 4.

c Is Equation 4 a multiple of Equation 1?

13 Consider the system of linear equations.

$$\text{Equation 1: } y = x + 8$$
$$\text{Equation 2: } y = -7x + 16$$

a Find the values of x and y that satisfy both equations.

b Add Equations 1 and 2 to create Equation 3. State Equation 3 in the form $y = mx + b$

c Multiply Equation 1 by the number 3 to create Equation 4. State Equation 4.

d Find the values of x and y that satisfy Equations 3 and 4.

e Are the solutions in the part (d) the same as for the system of Equations 1 and 2? Explain your reasoning.

14 Consider the equation $\frac{3x}{2} - 5 = \frac{2x}{3}$.

a Solve for the value of x that satisfies the equation.

b To verify the solution graphically, which two equations would need to be graphed?

- Equation 1: $y = \frac{2x}{3}$
- Equation 2: $y = 4$
- Equation 3: $y = 9x - 30$
- Equation 4: $y = \frac{3x}{2} - 5$

c Graph the equations in part (b) on the same plane.

d Hence, find the values of x that satisfies the two equations simultaneously.

15 Glen is looking for a mobile phone plan to s[redacted]p to. TeleNow has a plan that charges an initial fee of $55c$ per call, plus $3c$ per minute of calls. Konnex has a plan that charges an initial fee of $30c$ per call, plus $8c$ per minute of calls.
Let x be the duration of a call in minutes and y is the total cost for each phone call in cents.

a Complete table of values for each plan.

i TeleNow's plan

ii Konnex's plan

x	1	3	5	7	9
y					

b State whether each equation is the correct cost equation for TeleNow and Konnex's plans.

i TeleNow: $y+x=55+3$, Konnex: $y+x=30+8$

ii TeleNow: $y=55+x$, Konnex: $y=30+x$

iii TeleNow: $y=3x+55$, Konnex: $y=8x+30$

iv TeleNow: $y=55x+3$, Konnex: $y=30x+8$

c Graph the equations for TeleNow and Konnex's plans on the same plane.

d Find the duration of a phone call that would cost Glen the same on either plan.

e If Glen usually makes phone calls of 6 minutes or longer, which plan should Glen choose?

16 Consider the system of linear equations.

$$x+2y=-1$$
$$-4x+4y=1$$

State whether each system of linear equations has the same solution as the one given.

a
$$3y=-12$$
$$x+2y=-1$$

b
$$12y=-3$$
$$x+2y=-1$$

c
$$2x=-4$$
$$x+2y=-1$$

d
$$2x=-4$$
$$-4x+4y=1$$

Additional questions

17 Solve each system of equations:

a
$$\begin{cases} 5x + y &= 4 \\ 45x + 4y &= 16 \end{cases}$$

b
$$\begin{cases} y &= x + 2 \\ x + 2y &= 16 \end{cases}$$

c
$$\begin{cases} 7x + y = 65 \\ 3x + y = 33 \end{cases}$$

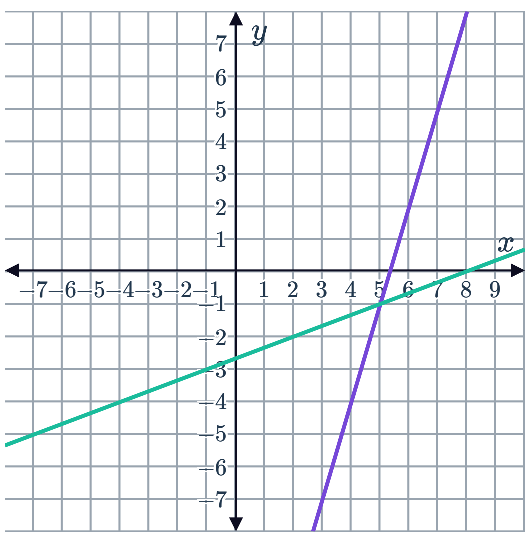
d
$$\begin{cases} x &= 5 - 2y \\ 3x + 4y &= 7 \end{cases}$$

e
$$\begin{cases} x + 4y &= 15 \\ -x + 3y &= -1 \end{cases}$$

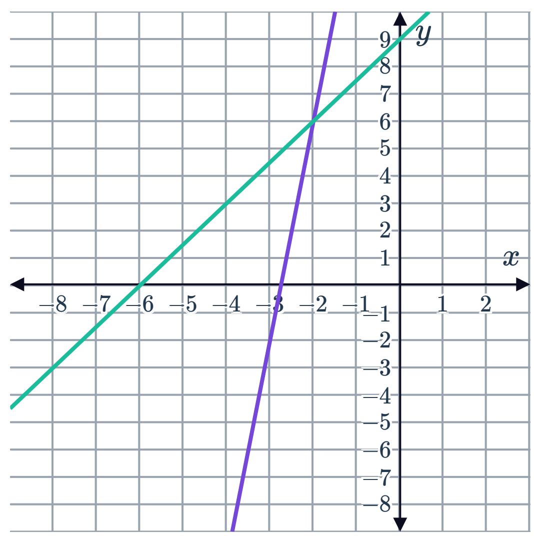
f
$$\begin{cases} 5x - 7y &= 3 \\ -3x - 7y &= -69 \end{cases}$$

18 Solve each system of equations for their point of intersection:

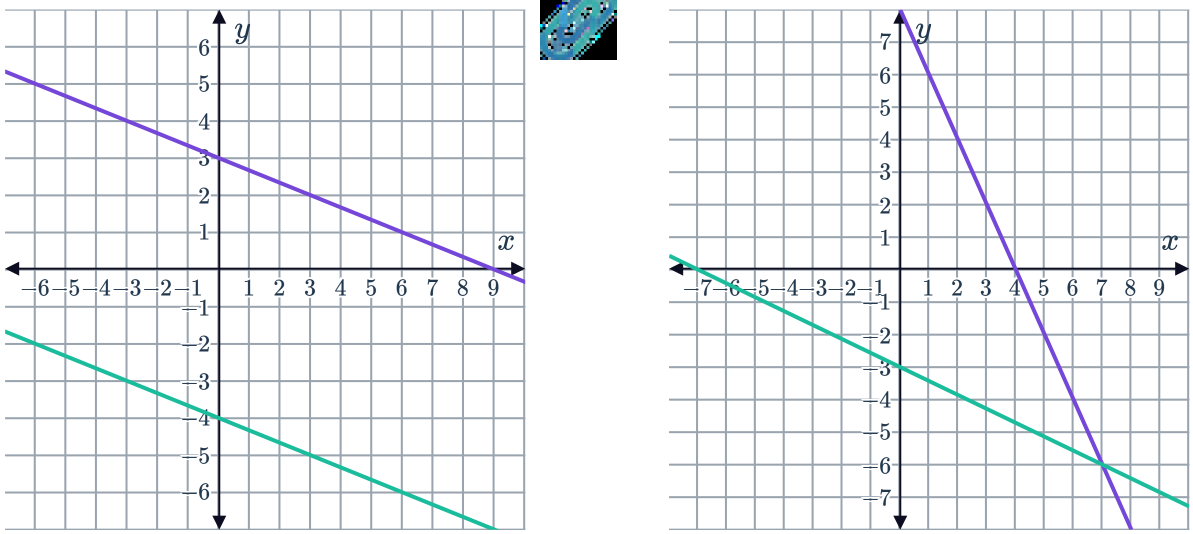
a



b



c



19 Solve each system of equations:

a

$$\begin{cases} y = 4.5x + 22 \\ 2.5x + 3y = 16 \end{cases}$$

b

$$\begin{cases} -6x - 38y = 2 \\ -2x - 19y = 34 \end{cases}$$

c

$$\begin{cases} y = 101 - 15x \\ y = -3x + 49 \end{cases}$$

d

$$\begin{cases} x = 75y - 400 \\ x = 38y + 53 \end{cases}$$